Excel Assignment 2 – Data Cleaning and Preparation

Data Analytics (DA) | Module 1 – Excel

Dataset	Product Dataset (Product ID, Product Name, Brand Name, Quantity, Category, Price)
Raw Data Rows	34 rows (before cleaning)
Cleaned Rows	31 rows (3 duplicates removed)
Issues Found	Missing prices (3), Missing categories (5), Name inconsistencies (3), Category typos (6), Duplic

Task 1 – Handling Missing Values

Step 1a: Checking for Missing Prices

I used Excel's Go To Special → Blanks feature to locate all empty cells in the Price column. 3 rows had missing price values. To handle these, I calculated the average price for each product's category and filled in the missing values with that average. This way I don't lose the rows but still have a reasonable price estimate.

Product ID	Product	Brand	Price Before	Qty	Category	Price After (imputed)
07-JUN-UK	Headphones	Sony	(blank)	45	Electroni → Electronics	$517.14 (Electronics avg)
14-AUG-US	Camping Tent	Coleman	(blank)	10	Outdoor	$130.00 (Outdoor avg)
03-JUN-CA	Sunglasses	Ray-Ban	(blank)	25	Fashion	$68.57 (Fashion avg)

Screenshot: Missing Price cells identified and filled with category average

Step 1b: Handling Missing Categories

5 rows had blank Category values. I looked up each product name and assigned the correct category manually based on what the product is. For example Backpack goes to Outdoor, Sneakers to Fashion, Coffee Maker to Kitchen, and Fitness Tracker to Electronics.

Product ID	Product Name	Brand	Category Before	Category After
22-MAY-US	Backpack	North Face	(blank)	Outdoor
25-NOV-AU	Sneakers	Adidas	(blank)	Fashion
08-DEC-DE	Coffee Maker	Nespresso	(blank)	Kitchen
19-JUL-BR	Fitness Tracker	Xiaomi	(blank)	Electronics

Screenshot: Missing Category values filled by product-name lookup

Task 2 – Correcting Inconsistent Data

Step 2a: Product Name Inconsistencies

Some product names were typed in all lowercase (e.g. 'laptop', 'smartphone', 'headphones') while the rest were in Title Case. I used the PROPER() function in the Workout sheet to standardise all names to Title Case, then used Find and Replace to fix the originals.

Before (inconsistent)	After (standardised)	Action
laptop	Laptop	Corrected to Title Case
smartphone	Smartphone	Corrected to Title Case
headphones	Headphones	Corrected to Title Case
T-shirt	T-Shirt	Corrected to Title Case

Screenshot: Product Name column after PROPER() function applied

Step 2b: Category Typos – Find and Replace

The category 'Electronics' was misspelled as 'Electroni' in 6 rows. I used Excel's Find and Replace (Ctrl+H) to fix all occurrences at once.

Find & Replace – Screenshot	
Find what:	Electroni
Replace with:	Electronics
Match entire cell contents:	Unchecked
Replacements made:	6

Screenshot: Find and Replace dialog used to fix 'Electroni' → 'Electronics'

Task 3 – Removing Duplicate Records

I used Excel's Data → Remove Duplicates feature, selecting all columns to check for exact row matches. 3 duplicate rows were found and removed. The dataset went from 34 rows to 31 rows.

Remove Duplicates – Screenshot		
Duplicate Row	**Product ID**	**Reason**
Row 24	21-AUG-CA (Laptop Bag, Samsonite)	Exact copy of Row 16
Row 28	16-APR-ES (Headphones, Bose)	Exact copy of Row 15
Row 33	17-JUN-IN (Laptop, HP)	Exact copy of Row 11

Screenshot: Remove Duplicates result – '3 duplicate values found and removed; 31 unique values remain'

Result	Dataset reduced from 34 rows → 31 rows after duplicate removal.

Task 4 – Splitting and Merging Columns

Step 4a: Splitting the Product ID column

The Product ID column contained date and country info combined (e.g. '28-JAN-US'). I used Text to Columns (Data → Text to Columns → Delimited → Delimiter: '-') to split it, then used RIGHT() to extract the country code and LEFT() to get the date part. I removed the original Product ID column after extraction.

Original Product ID	Manufacturing Date	Country Code	Formula (Date)	Formula (Country)
28-JAN-US	28-JAN	US	=LEFT(A2,6)	=RIGHT(A2,2)
07-JUN-UK	07-JUN	UK	=LEFT(A7,6)	=RIGHT(A7,2)
17-JUN-IN	17-JUN	IN	=LEFT(A11,6)	=RIGHT(A11,2)
14-MAY-RU	14-MAY	RU	=LEFT(A21,6)	=RIGHT(A21,2)
09-JUL-FR	09-JUL	FR	=LEFT(A34,6)	=RIGHT(A34,2)

Screenshot: Product ID column split into Manufacturing Date and Country Code

Step 4b: Merging Brand Name and Product Name

I created a new column called "Brand product" by combining Brand Name and Product Name using the & operator: =B2&" "&C2.; This gives a clean combined label for each product.

Brand Name	Product Name	Formula Used	Brand product (result)
Dell	Laptop	=B2&" "&C2	Laptop Dell
Samsung	Smartphone	=B4&" "&C4	Smartphone Samsung
Sony	Headphones	=B6&" "&C6	Headphones Sony
Apple	Tablet	=B9&" "&C9	Tablet Apple
Nikon	Camera	=B21&" "&C21	Camera Nikon

Screenshot: New "Brand product" column created by merging Brand + Product

Task 5 – Number Formatting

Step 5a: Price Column – Currency Format

I selected the entire Price column, right-clicked → Format Cells → Number → Currency, and set the format to $#,##0.00 (2 decimal places, dollar sign). This makes all prices display consistently as currency values.

Before (plain number)	After (currency format)
1000	$1,000.00
80	$80.00
900	$900.00
130	$130.00
500	$500.00
700	$700.00

Screenshot: Price column formatted as Currency ($#,##0.00)

Step 5b: Manufacturing Date – DD-MM-YYYY Format

After splitting the Product ID, the date part (e.g. '28-JAN') was combined with the month number using a VLOOKUP table and the year 2024 was added. The final format is DD-MM-YYYY.

Date from Product ID	Formatted Date (DD-MM-YYYY)
28-JAN	28-01-2024
07-JUN	07-06-2024
03-MAR	03-03-2024
14-OCT	14-10-2024
09-JUL	09-07-2024

Screenshot: Manufacturing Date column formatted as DD-MM-YYYY

Task 6 – Conditional Formatting

Step 6a: Data Bars on Price Column

I selected the Price column (E2:E32), went to Home → Conditional Formatting → Data Bars and chose the blue gradient fill. This creates a visual bar inside each cell that shows the price relative to all other prices — longer bar means higher price.

Product	Price	Data Bar (visual)
Laptop (Dell)	$1,000.00	■■■■■■■■■■■■■■■■■■■■ (longest)
Camera (Nikon)	$700.00	■■■■■■■■■■■■■■
Tablet (Apple)	$500.00	■■■■■■■■■■
Headphones (Bose)	$250.00	■■■■■
Blender (Ninja)	$90.00	■■
Toaster (Hamilton)	$40.00	■ (shortest)

Screenshot: Data bars applied to Price column – blue gradient fill, auto-scaled

Step 6b: Highlighting 'Electronics' Category

I selected the Category column (G2:G32), went to Home → Conditional Formatting → New Rule → 'Format only cells that contain' → Cell Value equals 'Electronics'. I set the fill to yellow (#FFD700) and font to bold. This makes all Electronics products stand out immediately.

Product Name	Category	Highlight Applied?
Laptop	**Electronics**	**Yes – Gold background, Bold**
Smartphone	**Electronics**	**Yes – Gold background, Bold**
Tablet	**Electronics**	**Yes – Gold background, Bold**
Headphones	**Electronics**	**Yes – Gold background, Bold**
Sneakers	Fashion	No highlight
Coffee Maker	Kitchen	No highlight
Hiking Boots	Outdoor	No highlight

Screenshot: Category column – Electronics rows highlighted in gold with bold text

Summary

All 6 cleaning tasks have been completed on the Product Dataset. The dataset went from 34 raw rows with multiple issues to 31 fully cleaned rows ready for analysis. Below is a summary of all steps performed:

Task	Issue	Action Taken	Done
1a	Missing Price (3 rows)	Imputed with category average price	✓
1b	Missing Category (5 rows)	Assigned based on product name	✓
2a	Product Name lowercase (3)	Fixed using PROPER() + Find & Replace	✓
2b	Category typo 'Electroni' (6)	Fixed using Find & Replace	✓
3	Duplicate rows (3)	Removed using Remove Duplicates	✓
4a	Product ID split	Extracted Date + Country Code columns	✓
4b	Brand + Name merge	New 'Brand product' column created	✓
5a	Price formatting	Applied $#,##0.00 currency format	✓
5b	Date formatting	Displayed as DD-MM-YYYY	✓
6a	Data Bars on Price	Blue gradient data bars added	✓
6b	Electronics highlight	Gold + bold for Electronics category	✓